

total, 59.2% of questions can be answered based on knowledge after 2020, suggesting that there is still room for improvement to let temporal-adaptive LMs elicit more recent knowledge from finetuning. We leave this as one of the problems worth investigating in the future.

Question	Target-year Ft. Answer	Temporal-adaptive Ft. Answer
What is the title of the show where Taraji P. Henson appeared in her latest television role?	The Best of the Oscars	Empire (2015-2020)
Who is the current shirt sponsor of Sevilla FC?	Codere	Marathonbet (2019-2021)
What is the most recent film that Matthew Lillard has lent his voice to?	Teenage Mutant Ninja Turtles: Mutant Mayhem	Trick or Treat Scooby-Doo! (2022)
In which city was the International Philosophy Olympiad last held?	Sarajevo	Oslo (2021)
Who is the captain of the winning team in the last UEFA Champions League final?	Thiago	Iker Casillas (2014)

Table 23: Sampled questions that cannot be answered correctly as of the knowledge in 2022 by the target-year finetuning model but can be responded with historically correct answers by the temporal-adaptive finetuned model. The years that the temporal-adaptive finetuned LM’s answers being correct are shown in the parenthesis.

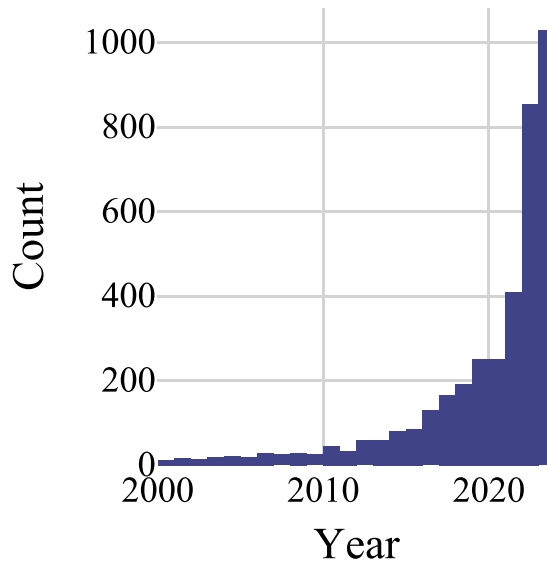


Figure 8: The distribution of the most recent year of the answers given by the temporal-adaptive LM to be correct.

F Detailed F1 scores of unaligned and aligned LMs

To provide detailed information, we demonstrate the detailed target-year F1 scores of the LMs presented in Figure 1 regarding every year between 2000 to 2023 in Table 24.

Model ↓ Year →	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Davinci-002	6.4	6.7	6.8	7.0	7.3	8.0	8.3	8.7	9.4	9.6	10.2	11.5	12.3	13.8	14.8	15.1	15.6	16.6	16.4	15.0	12.3	11.4	10.0	9.0
LLaMA1-65B	6.7	7.2	7.3	7.7	7.7	8.9	9.4	10.3	10.8	11.5	12.0	13.7	15.0	16.7	18.1	19.2	19.3	20.0	20.1	18.8	14.3	13.2	12.1	10.3
LLaMA2-70B	6.4	6.5	6.6	7.1	7.0	8.0	7.9	9.0	9.1	9.3	10.2	11.1	12.0	13.7	15.0	16.6	17.4	19.7	21.5	23.0	20.6	19.8	17.2	14.3
LLaMA2-7B	5.7	6.0	6.1	6.4	6.5	7.1	7.4	7.9	8.0	8.6	8.9	9.5	10.1	11.4	12.1	12.9	13.2	13.6	13.9	14.1	12.0	11.4	10.4	9.4
ChatGPT	5.2	5.4	5.2	5.5	5.6	5.9	6.1	6.6	6.9	7.1	7.6	8.1	8.6	9.9	10.5	11.8	12.5	14.1	15.8	19.4	23.0	23.8	17.1	14.2
Tulu2	5.6	5.9	5.9	6.4	6.4	7.0	7.3	7.8	8.0	8.1	8.6	9.3	10.0	11.4	12.7	13.6	14.5	16.3	18.7	22.9	22.5	19.1	16.2	13.5
LLaMA2-70B Fine-tuned	6.2	6.6	6.7	7.1	7.3	7.8	8.1	8.3	8.5	8.6	9.3	9.9	10.4	11.6	12.3	13.6	14.3	15.5	16.9	19.1	19.8	23.7	27.9	19.1
LLaMA2-70B Temporal adaptive	6.7	6.9	7.1	7.6	7.8	8.1	8.6	9.1	9.3	9.5	10.0	10.8	11.5	12.9	13.9	15.4	16.2	17.8	19.0	20.7	19.9	23.5	25.7	17.6

Table 24: The target-year F1 score $\mathcal{F}^y$ regarding every year between 2000 and 2023 on models demonstrated in Figure 1. The maximum F1 score of each LM is bolded.